IEEE eSCIENCE 2026 WORKSHOP

AGENT4SC 2026

1st Workshop on Agentic AI for Large-scale Science

Important Dates

Papers Due	**July 21, 2026**
Notification of Acceptance	**August 3, 2026**
Camera-Ready Papers	**August 10, 2026**
Workshop	**IEEE eScience 2026**

Submit via **EasyChair**: https://easychair.org/my/conference?conf=escience2026

Select track: ***1st Workshop on Agentic AI for Large-scale Science***

About AGENT4SC

Agentic AI is rapidly moving from "chat with tools" prototypes to autonomous systems that can reason, plan, coordinate, and act across complex digital research ecosystems. For the eScience community, this represents the emergence of a new **control plane** for computational and data-driven research.

Agents can request allocations, launch ensembles, steer workflows, move PB-scale datasets, trigger experiment/compute co-scheduling, and generate decisions that impact scientific validity—spanning the computing continuum from instruments and autonomous laboratories to simulations, data centers, cloud systems, and leadership-class HPC.

AGENT4SC provides a dedicated forum to advance **scalable architectures, cross-continuum coordination, evaluation frameworks, safety and verification strategies, provenance, and observability mechanisms** before agentic systems become embedded in production research workflows.

Topics of Interest

- System architectures & design principles for agentic AI at scale
- Agentic AI for the edge–cloud–HPC continuum
- Agent memory & long-horizon reasoning under compute constraints
- Data & metadata architectures (vector stores, knowledge graphs)
- Human-in-the-loop & oversight for long autonomous campaigns
- Provenance, observability, auditability, and reproducibility
- Safety, security & accountability in production science
- Hallucination detection, mitigation, and recovery
- Performance modeling under realistic scientific workloads
- Interoperability & standardization (interfaces, protocols, schemas)
- Cross-platform scheduling under agent-driven control
- Real-world experience reports from large-scale deployments

Paper Formats

Pages	Type
4	**Short Paper** — work-in-progress (*not including references*)
8	**Full Paper** — complete research (*not including references*)

Submission Guidelines

- Papers must be written in **English** and formatted per IEEE eScience 2026 author guidelines
- All papers receive **at least three** peer reviews from the Program Committee
- Accepted papers published in the official **eScience 2026 workshop proceedings**
- Authors are encouraged to include reproducibility information about software, data artifacts, and AI assistants used

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